

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

1. If $\sin x + \sin^2 x = 1$, then the value of $\cos^2 x + \cos^4 x$ is -
 (A) 0 (B) 2 (C) 1 (D) 3 MTR001
2. The value of $\sin 10^\circ + \sin 20^\circ + \sin 30^\circ + \dots + \sin 360^\circ$ is equal to -
 (A) 0 (B) 1 (C) $\sqrt{3}$ (D) 2 MTR002
3. If $\cos x + \cos y + \cos \alpha = 0$ and $\sin x + \sin y + \sin \alpha = 0$, then $\cot\left(\frac{x+y}{2}\right) =$
 (A) $\sin \alpha$ (B) $\cos \alpha$ (C) $\cot \alpha$ (D) $2 \sin \alpha$ MTR003
4. $\frac{\sin(A-C) + 2\sin A + \sin(A+C)}{\sin(B-C) + 2\sin B + \sin(B+C)}$ is equal to -
 (A) $\tan A$ (B) $\frac{\sin A}{\sin B}$ (C) $\frac{\cos A}{\cos B}$ (D) $\frac{\sin C}{\cos B}$ MTR004
5. The expression $\frac{\sin 8\theta \cos \theta - \sin 6\theta \cos 3\theta}{\cos 2\theta \cos \theta - \sin 3\theta \sin 4\theta}$ is equals -
 (A) $\tan \theta$ (B) $\tan 2\theta$ (C) $\sin 2\theta$ (D) $\cos 2\theta$ MTR005
6. If $\tan \alpha = (1 + 2^{-x})^{-1}$, $\tan \beta = (1 + 2^{x+1})^{-1}$, then $\alpha + \beta =$
 (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$ MTR009
7. If $\tan B = \frac{n \sin A \cos A}{1 - n \cos^2 A}$ then $\tan(A+B)$ equals
 (A) $\frac{\sin A}{(1-n)\cos A}$ (B) $\frac{(n-1)\cos A}{\sin A}$ (C) $\frac{\sin A}{(n-1)\cos A}$ (D) $\frac{\sin A}{(n+1)\cos A}$ MTR010
8. $\frac{1 + \sin 2\theta + \cos 2\theta}{1 + \sin 2\theta - \cos 2\theta} =$
 (A) $\frac{1}{2} \tan \theta$ (B) $\frac{1}{2} \cot \theta$ (C) $\tan \theta$ (D) $\cot \theta$ MTR006
9. If $A = \tan 6^\circ \tan 42^\circ$ and $B = \cot 66^\circ \cot 78^\circ$, then -
 (A) $A = 2B$ (B) $A = 1/3B$ (C) $A = B$ (D) $3A = 2B$ MTR007
10. In a right angled triangle, the hypotenuse is $2\sqrt{2}$ times the perpendicular drawn from the opposite vertex. Then the other acute angles of the triangle are
 (A) $\frac{\pi}{3}$ and $\frac{\pi}{6}$ (B) $\frac{\pi}{8}$ and $\frac{3\pi}{8}$ (C) $\frac{\pi}{4}$ and $\frac{\pi}{4}$ (D) $\frac{\pi}{5}$ and $\frac{3\pi}{10}$ MTR008

11. The value of $\operatorname{cosec} \frac{\pi}{18} - \sqrt{3} \sec \frac{\pi}{18}$ is a
 (A) surd (B) rational which is not integral
 (C) negative integer (D) natural number MTR011
12. The value of $\cot x + \cot(60^\circ + x) + \cot(120^\circ + x)$ is equal to :
 (A) $\cot 3x$ (B) $\tan 3x$ (C) $3 \tan 3x$ (D) $\frac{3 - 9 \tan^2 x}{3 \tan x - \tan^3 x}$ MTR012
13. If $\frac{5\pi}{2} < x < 3\pi$, then the value of the expression $\frac{\sqrt{1 - \sin x} + \sqrt{1 + \sin x}}{\sqrt{1 - \sin x} - \sqrt{1 + \sin x}}$ is
 (A) $-\cot \frac{x}{2}$ (B) $\cot \frac{x}{2}$ (C) $\tan \frac{x}{2}$ (D) $-\tan \frac{x}{2}$ MTR013
14. The value of $\cot 7\frac{1}{2}^\circ + \tan 67\frac{1}{2}^\circ - \cot 67\frac{1}{2}^\circ - \tan 7\frac{1}{2}^\circ$ is :
 (A) a rational number (B) irrational number
 (C) $2(3 + 2\sqrt{3})$ (D) $2(3 - \sqrt{3})$ MTR014
15. If $x + y = 3 - \cos 4\theta$ and $x - y = 4 \sin 2\theta$ then
 (A) $x^4 + y^4 = 9$ (B) $\sqrt{x} + \sqrt{y} = 16$
 (C) $x^3 + y^3 = 2(x^2 + y^2)$ (D) $\sqrt{x} + \sqrt{y} = 2$ MTR015
16. If $\tan A + \tan B + \tan C = \tan A \cdot \tan B \cdot \tan C$, then -
 (A) A, B, C must be angles of a triangle
 (B) the sum of any two of A, B, C is equal to the third
 (C) $A + B + C$ must be an integral multiple of π
 (D) None of these MTR016
17. If $f(x) = \frac{\sin 3x}{\sin x}$, $x \neq n\pi$, then the range of values of $f(x)$ for real values of x is -
 (A) $[-1, 3]$ (B) $(-\infty, -1]$ (C) $(3, +\infty)$ (D) $[-1, 3]$ MTR017
18. Maximum and minimum value of $2\sin^2 \theta - 3\sin \theta + 2$ is -
 (A) $\frac{1}{4}, -\frac{7}{4}$ (B) $\frac{1}{4}, \frac{21}{4}$ (C) $\frac{21}{4}, -\frac{3}{4}$ (D) $7, \frac{7}{8}$ MTR018
19. For $\theta \in (0, \pi/2)$, the maximum value of $\sin\left(\theta + \frac{\pi}{6}\right) + \cos\left(\theta + \frac{\pi}{6}\right)$ is attained at $\theta =$
 (A) $\frac{\pi}{12}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{4}$ MTR019
20. Minimum value of the expression $\cos^2 \theta - (6\sin \theta \cos \theta) + 3 \sin^2 \theta + 2$, is -
 (A) $4 + \sqrt{10}$ (B) $4 - \sqrt{10}$ (C) 0 (D) 4 MTR020

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. If $\cos x = \tan x$, then which of the following is/are true?
 (A) $\frac{1}{\sin x} + \cos^4 x = 1$ (B) $\frac{1}{\sin x} + \cos^4 x = 2$
 (C) $\cos^4 x + \cos^2 x = 1$ (D) $\cos^4 x + \cos^2 x = 2$ MTR045
2. If $\cos(A - B) = \frac{3}{5}$ and $\tan A \tan B = 2$, then which of the following is/are correct?
 (A) $\cos A \cos B = -\frac{1}{5}$ (B) $\sin A \sin B = \frac{2}{5}$ (C) $\cos(A + B) = -\frac{1}{5}$ (D) $\sin A \cos B = \frac{4}{5}$ MTR046
3. If $3\sin\beta = \sin(2\alpha + \beta)$, then $\tan(\alpha + \beta) - 2\tan\alpha$ is
 (A) independent of α
 (B) independent of β
 (C) dependent of both α and β
 (D) independent of α but dependent of β MTR048
4. If $x + y = z$, then $\cos^2 x + \cos^2 y + \cos^2 z - 2\cos x \cos y \cos z$ is equal to
 (A) $\cos^2 z$ (B) $\sin^2 z$ (C) $\cos(x + y - z)$ (D) 1 MTR051
5. If $L = \cos^2 84^\circ + \cos^2 36^\circ + \cos 36^\circ \cos 84^\circ$
 $M = \cot 73^\circ \cot 47^\circ \cot 13^\circ$
 $N = 4\sin 156^\circ \sin 84^\circ \sin 36^\circ$, then which of the following option(s) is(are) correct?
 (A) $L < 1$ (B) $M > \tan 2$ (C) $N > \sin \frac{\pi}{4}$ (D) $0 < LMN$ MTR049
6. If $P_n = \cos^n \theta + \sin^n \theta$ and $Q_n = \cos^n \theta - \sin^n \theta$, then which of the following is/are true.
 (A) $P_n - P_{n-2} = -\sin^2 \theta \cos^2 \theta P_{n-4}$ (B) $Q_n - Q_{n-2} = -\sin^2 \theta \cos^2 \theta Q_{n-4}$
 (C) $P_4 = 1 - 2\sin^2 \theta \cos^2 \theta$ (D) $Q_4 = \cos^2 \theta - \sin^2 \theta$ MTR054
7. If $\alpha + \beta + \gamma = 2\pi$, then
 (A) $\tan \frac{\alpha}{2} + \tan \frac{\beta}{2} + \tan \frac{\gamma}{2} = \tan \frac{\alpha}{2} \tan \frac{\beta}{2} \tan \frac{\gamma}{2}$
 (B) $\tan \frac{\alpha}{2} \tan \frac{\beta}{2} + \tan \frac{\beta}{2} \tan \frac{\gamma}{2} + \tan \frac{\gamma}{2} \tan \frac{\alpha}{2} = 1$
 (C) $\tan \frac{\alpha}{2} + \tan \frac{\beta}{2} + \tan \frac{\gamma}{2} = -\tan \frac{\alpha}{2} \tan \frac{\beta}{2} \tan \frac{\gamma}{2}$
 (D) $\tan \frac{\alpha}{4} \tan \frac{\beta}{4} + \tan \frac{\beta}{4} \tan \frac{\gamma}{4} + \tan \frac{\gamma}{4} \tan \frac{\alpha}{4} = 1$ MTR050

8. Let $y = \frac{\cos x + \cos 2x + \cos 3x + \cos 4x + \cos 5x + \cos 6x + \cos 7x}{\sin x + \sin 2x + \sin 3x + \sin 4x + \sin 5x + \sin 6x + \sin 7x}$, then which of the following hold good?
- (A) The value of y when $x = \pi/8$ is not defined.
 (B) The value of y when $x = \pi/16$ is 1.
 (C) The value of y when $x = \pi/32$ is $\sqrt{2} - 1$.
 (D) The value of y when $x = \pi/48$ is $2 + \sqrt{3}$.

MTR053

9. Let $f_n(\theta) = \sum_{r=0}^n \frac{1}{4^r} \sin^4(2^r \theta)$. Then which of the following alternative(s) is/are correct ?
- (A) $f_2\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$ (B) $f_3\left(\frac{\pi}{8}\right) = \frac{2+\sqrt{2}}{4}$ (C) $f_4\left(\frac{3\pi}{2}\right) = 1$ (D) $f_5(\pi) = 0$

MTR052

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 10 to 12

Consider the polynomial $P(x) = (x - \cos 36^\circ)(x - \cos 84^\circ)(x - \cos 156^\circ)$

10. The coefficient of x^2 is
 (A) 0 (B) 1 (C) $-\frac{1}{2}$ (D) $\frac{\sqrt{5}-1}{2}$
11. The coefficient of x is
 (A) $\frac{3}{2}$ (B) $-\frac{3}{2}$ (C) $-\frac{3}{4}$ (D) zero
12. The absolute term in $P(x)$ has the value equal to
 (A) $\frac{\sqrt{5}-1}{4}$ (B) $\frac{\sqrt{5}-1}{16}$ (C) $\frac{\sqrt{5}+1}{16}$ (D) $\frac{1}{16}$

MTR055

MTR056

MTR057

MATCHING LIST TYPE QUESTION

- | 13. | List-I | List-II |
|-------|--|-------------------------|
| (I) | The value of $\tan 20^\circ \cdot \tan 40^\circ \cdot \tan 60^\circ \cdot \tan 80^\circ$ is | (P) Odd natural Number |
| (II) | The value of $4\left(\sin^4 \frac{\pi}{16} + \sin^4 \frac{3\pi}{16} + \sin^4 \frac{5\pi}{16} + \sin^4 \frac{7\pi}{16}\right)$ is | (Q) Even natural Number |
| (III) | $\tan \alpha + 2 \tan 2\alpha + 4 \tan 4\alpha + 8 \cot 8\alpha = \lambda \cot \lambda \alpha$ then λ is | (R) 3 |
| (IV) | $(4 \cos^2 9^\circ - 3)(4 \cos^2 27^\circ - 3) = \tan k^\circ$, then k is | (S) 6 |
| | | (T) 1 |
| | | (U) 9 |

Which of the following is **CORRECT** ?

- (A) (I) $\rightarrow$ (P,R) ; (II) $\rightarrow$ (Q,S) ; (III) $\rightarrow$ (P,T) ; (IV) $\rightarrow$ (P, U)
 (B) (I) $\rightarrow$ (P,T) ; (II) $\rightarrow$ (Q,R) ; (III) $\rightarrow$ (P,U) ; (IV) $\rightarrow$ (R, U)
 (C) (I) $\rightarrow$ (R,S) ; (II) $\rightarrow$ (T,U) ; (III) $\rightarrow$ (P,T) ; (IV) $\rightarrow$ (P, Q)
 (D) (I) $\rightarrow$ (P,R) ; (II) $\rightarrow$ (P,U) ; (III) $\rightarrow$ (Q,S) ; (IV) $\rightarrow$ (S, T)

MTR058

MATRIX MATCH TYPE QUESTION

14. In the following matrix match **Column-I** has some quantities and **Column-II** has some comments or other quantities.

Match the each element in **Column-I** with corresponding element(s) in **Column-II**

Column-I		Column-II	
(A)	The value of $\left 4 \left(2 \cos^3 \frac{\pi}{7} - \cos^2 \frac{\pi}{7} - \cos \frac{\pi}{7} \right) \right $ is	(P)	4
(B)	If $A + B + C = \pi$ and $\cos A = \cos B \cdot \cos C$ then $\tan B \cdot \tan C$ has the value equal to	(Q)	8
(C)	$4 \left(\frac{\cos 20^\circ + 8 \sin 70^\circ \sin 50^\circ \sin 10^\circ}{\sin^2 80^\circ} \right)$ is equal to	(R)	2
(D)	The maximum value of $12 \sin \theta - 9 \sin^2 \theta$ is	(S)	1
		(T)	6

MTR059

EXERCISE (S)

1. $2(\sin^6 \theta + \cos^6 \theta) - 3(\sin^4 \theta + \cos^4 \theta) + 1$ is equal to - MTR021
2. If $x = y \cos \frac{2\pi}{3} = z \cos \frac{4\pi}{3}$ then $xy + yz + zx =$ MTR022
3. If $\cos(\alpha + \beta) + \sin(\alpha - \beta) = 0$ and $2010 \tan \beta + 1 = 0$, then $|\tan \alpha|$ is equal to MTR023
4. Calculate: $\frac{2\cos 40^\circ - \cos 20^\circ}{\sin 20^\circ}$ MTR064
5. If acute angle $A = 3B$ and $\sin A = \frac{4}{5}$ then $\left| \frac{3\sec B - 4\operatorname{cosec} B}{2} \right|$ is MTR060
6. If $3\sin \alpha = 5\sin \beta$, then $\frac{\tan \frac{\alpha + \beta}{2}}{\tan \frac{\alpha - \beta}{2}} =$ MTR025
7. Given that $(1 + \tan 1^\circ)(1 + \tan 2^\circ) \dots (1 + \tan 45^\circ) = 2^n$, find $\left(\frac{n}{4}\right)$. MTR063
8. If the value of the expression $\sin 25^\circ \cdot \sin 35^\circ \cdot \sin 85^\circ$ can be expressed as $\frac{\sqrt{a} + \sqrt{b}}{c}$, where $a, b, c \in N$ and are in their lowest form, find the value of $\left(\frac{a+b+c}{18}\right)$. MTR062
9. If $\tan x + \tan y = 25$ and $\cot x + \cot y = 30$, then the value of $\tan(x + y)$ is MTR024
10. If $\cos \alpha = \frac{2\cos \beta - 1}{2 - \cos \beta}$ then $\tan^2 \frac{\alpha}{2} \cdot \cot^2 \frac{\beta}{2}$ has the value equal to {where $\alpha, \beta \in (0, \pi)$ } MTR026
11. If the value of $\sin \frac{\pi}{14} \sin \frac{3\pi}{14} \sin \frac{5\pi}{14}$ is $\frac{p}{q}$ (in simplest form) then what is the value of $p + q$? MTR028
12. The exact value of $\frac{96 \sin 80^\circ \sin 65^\circ \sin 35^\circ}{\sin 20^\circ + \sin 50^\circ + \sin 110^\circ}$ is equal to MTR029

13. If m and n are positive integers satisfying

$$1 + \cos 2\theta + \cos 4\theta + \cos 6\theta + \cos 8\theta + \cos 10\theta = \frac{\cos m\theta \cdot \sin n\theta}{\sin \theta} \text{ then } (m + n) \text{ is equal to}$$

MTR030

14. If $A = \sin \frac{2\pi}{7} + \sin \frac{4\pi}{7} + \sin \frac{8\pi}{7}$ and $B = \cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{8\pi}{7}$ then $A^2 + B^2$ is equal to

MTR027

15. If $k_1 = \tan 27\theta - \tan \theta$ and $k_2 = \frac{\sin \theta}{\cos 3\theta} + \frac{\sin 3\theta}{\cos 9\theta} + \frac{\sin 9\theta}{\cos 27\theta}$ then $\left(\frac{k_1}{k_2} \right)$ is equal to

MTR061

EXERCISE - JEE (Main) PYQ

1. For any $\theta \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$, the expression $3(\sin\theta - \cos\theta)^4 + 6(\sin\theta + \cos\theta)^2 + 4\sin^6\theta$ equals :

[JEE (Main) 2019]

(1) $13 - 4\cos^6\theta$

(2) $13 - 4\cos^4\theta + 2\sin^2\theta\cos^2\theta$

(3) $13 - 4\cos^2\theta + 6\cos^4\theta$

(4) $13 - 4\cos^2\theta + 6\sin^2\theta\cos^2\theta$

MTR034
2. The value of $\cos\frac{\pi}{2^2} \cdot \cos\frac{\pi}{2^3} \cdot \dots \cdot \cos\frac{\pi}{2^{10}} \cdot \sin\frac{\pi}{2^{10}}$ is :

[JEE (Main) 2019]

(1) $\frac{1}{256}$

(2) $\frac{1}{2}$

(3) $\frac{1}{512}$

(4) $\frac{1}{1024}$

MTR035
3. Let $f_k(x) = \frac{1}{k}(\sin^k x + \cos^k x)$ for $k = 1, 2, 3, \dots$. Then for all $x \in R$, the value of $f_4(x) - f_6(x)$ is equal to :-

[JEE (Main) 2019]

(1) $\frac{5}{12}$

(2) $-\frac{1}{12}$

(3) $\frac{1}{4}$

(4) $\frac{1}{12}$

MTR036
4. The maximum value of $3\cos\theta + 5\sin\left(\theta - \frac{\pi}{6}\right)$ for any real value of θ is :

[JEE (Main) 2019]

(1) $\sqrt{19}$

(2) $\frac{\sqrt{79}}{2}$

(3) $\sqrt{31}$

(4) $\sqrt{34}$

MTR037
5. Let α and β be two real roots of the equation $(k+1)\tan^2 x - \sqrt{2}\lambda \tan x = (1-k)$, where $k \neq -1$ and λ are real numbers. If $\tan^2(\alpha + \beta) = 50$, then a value of λ is ;

[JEE (Main) 2020]

(1) 5

(2) 10

(3) $5\sqrt{2}$

(4) $10\sqrt{2}$

MTR038
6. The value of $\cos^3\left(\frac{\pi}{8}\right) \cdot \cos\left(\frac{3\pi}{8}\right) + \sin^3\left(\frac{\pi}{8}\right) \cdot \sin\left(\frac{3\pi}{8}\right)$ is :

[JEE (Main) 2020]

(1) $\frac{1}{4}$

(2) $\frac{1}{\sqrt{2}}$

(3) $\frac{1}{2\sqrt{2}}$

(4) $\frac{1}{2}$

MTR039
7. If $15\sin^4\alpha + 10\cos^4\alpha = 6$, for some $\alpha \in R$, then the value of $27\sec^6\alpha + 8\operatorname{cosec}^6\alpha$ is equal to :

[JEE (Main) 2021]

(1) 350

(2) 500

(3) 400

(4) 250

MTR040
8. $16\sin(20^\circ) \sin(40^\circ) \sin(80^\circ)$ is equal to:

[JEE (Main) 2022]

(1) $\sqrt{3}$

(2) $2\sqrt{3}$

(3) 3

(4) $4\sqrt{3}$

MTR041

Trigonometric Ratios and Identities

9. If $\cot \alpha = 1$ and $\sec \beta = -\frac{5}{3}$, where $\pi < \alpha < \frac{3\pi}{2}$ and $\frac{\pi}{2} < \beta < \pi$, then the value of $\tan(\alpha + \beta)$ and the quadrant in which $\alpha + \beta$ lies, respectively are [JEE (Main) 2022]

- (1) $-\frac{1}{7}$ and IVth quadrant (2) 7 and Ist quadrant
(3) -7 and IVth quadrant (4) $\frac{1}{7}$ and Ist quadrant

MTR042

10. The value of $\cos\left(\frac{2\pi}{7}\right) + \cos\left(\frac{4\pi}{7}\right) + \cos\left(\frac{6\pi}{7}\right)$ is equal to : [JEE (Main) 2022]

- (1) -1 (2) $-\frac{1}{2}$ (3) $-\frac{1}{3}$ (4) $-\frac{1}{4}$

MTR043

11. If $\sin^2(10^\circ)\sin(20^\circ)\sin(40^\circ)\sin(50^\circ)\sin(70^\circ) = \alpha - \frac{1}{16}\sin(10^\circ)$, then $16 + \alpha^{-1}$ is equal to _____. [JEE (Main) 2022]

MTR044

12. The value of $36(4\cos^2 9^\circ - 1)(4\cos^2 27^\circ - 1)(4\cos^2 81^\circ - 1)(4\cos^2 243^\circ - 1)$ is [JEE (Main) 2023]

- (1) 54 (2) 18 (3) 27 (4) 36

MTR074

13. If $\tan 15^\circ + \frac{1}{\tan 75^\circ} + \frac{1}{\tan 105^\circ} + \tan 195^\circ = 2a$, then the value of $\left(a + \frac{1}{a}\right)$ is : [JEE (Main) 2023]

- (1) 4 (2) $4 - 2\sqrt{3}$ (3) 2 (4) $5 - \frac{3}{2}\sqrt{3}$

MTR075

14. $96\cos\frac{\pi}{33}\cos\frac{2\pi}{33}\cos\frac{4\pi}{33}\cos\frac{8\pi}{33}\cos\frac{16\pi}{33}$ is equal to [JEE (Main) 2023]

- (1) 3 (2) 2 (3) 4 (4) 1

MTR076

15. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = \log_{\sqrt{m}}\left\{\sqrt{2}(\sin x - \cos x) + m - 2\right\}$, for some m , such that the range of f is $[0, 2]$. Then the value of m is. [JEE (Main) 2023]

- (1) 5 (2) 3 (3) 2 (4) 4

MTR077

16. Let the set of all $a \in \mathbb{R}$ such that the equation $\cos 2x + a \sin x = 2a - 7$ has a solution be $[p, q]$ and $r = \tan 9^\circ - \tan 27^\circ - \frac{1}{\cot 63^\circ} + \tan 81^\circ$, then pqr is equal to _____. [JEE (Main) 2024]

MTR126

17. For $\alpha, \beta \in \left(0, \frac{\pi}{2}\right)$, let $3\sin(\alpha + \beta) = 2\sin(\alpha - \beta)$ and a real number k be such that $\tan \alpha = k \tan \beta$. Then the value of k is equal to : [JEE (Main) 2024]

- (1) $-\frac{2}{3}$ (2) -5 (3) $\frac{2}{3}$ (4) 5

MTR127

18. Let a rectangle $ABCD$ of sides 2 and 4 be inscribed in another rectangle $PQRS$ such that the vertices of the rectangle $ABCD$ lie on the sides of the rectangle $PQRS$. Let a and b be the sides of the rectangle $PQRS$ when its area is maximum. Then $(a + b)^2$ is equal to : [JEE (Main) 2024]

- (1) 72 (2) 60 (3) 80 (4) 64

MTR128

19. If $\tan A = \frac{1}{\sqrt{x(x^2 + x + 1)}}$, $\tan B = \frac{\sqrt{x}}{\sqrt{x^2 + x + 1}}$ and $\tan C = (x^{-3} + x^{-2} + x^{-1})^{\frac{1}{2}}$, $0 < A, B, C < \frac{\pi}{2}$, then

$A + B$ is equal to :

[JEE (Main) 2024]

- (1) C (2) $\pi - C$ (3) $2\pi - C$ (4) $\frac{\pi}{2} - C$

MTR129

EXERCISE - JEE (Advanced) PYQ

1. Let $\theta \in \left(0, \frac{\pi}{4}\right)$ and $t_1 = (\tan \theta)^{\tan \theta}$, $t_2 = (\tan \theta)^{\cot \theta}$, $t_3 = (\cot \theta)^{\tan \theta}$, $t_4 = (\cot \theta)^{\cot \theta}$, then -

[JEE (Advanced) 2006]

(A) $t_1 > t_2 > t_3 > t_4$

(B) $t_4 > t_3 > t_1 > t_2$

(C) $t_3 > t_1 > t_2 > t_4$

(D) $t_2 > t_3 > t_1 > t_4$

MTR065

2. If $\frac{\sin^4 x}{2} + \frac{\cos^4 x}{3} = \frac{1}{5}$, then

[JEE (Advanced) 2009]

(A) $\tan^2 x = \frac{2}{3}$

(B) $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{1}{125}$

(C) $\tan^2 x = \frac{1}{3}$

(D) $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{2}{125}$

MTR066

3. For $0 < \theta < \frac{\pi}{2}$, the solution(s) of $\sum_{m=1}^6 \operatorname{cosec}\left(\theta + \frac{(m-1)\pi}{4}\right) \operatorname{cosec}\left(\theta + \frac{m\pi}{4}\right) = 4\sqrt{2}$ is (are) -

[JEE (Advanced) 2009]

(A) $\frac{\pi}{4}$

(B) $\frac{\pi}{6}$

(C) $\frac{\pi}{12}$

(D) $\frac{5\pi}{12}$

MTR067

4. The maximum value of the expression $\frac{1}{\sin^2 \theta + 3 \sin \theta \cos \theta + 5 \cos^2 \theta}$ is

[JEE (Advanced) 2010]

MTR068

5. Let $P = \{\theta : \sin \theta - \cos \theta = \sqrt{2} \cos \theta\}$ and $Q = \{\theta : \sin \theta + \cos \theta = \sqrt{2} \sin \theta\}$ be two sets. Then

[JEE (Advanced) 2013]

(A) $P \subset Q$ and $Q - P \neq \emptyset$

(B) $Q \not\subset P$

(C) $P \not\subset Q$

(D) $P = Q$

MTR070

6. The value of $\sum_{k=1}^{13} \frac{1}{\sin\left(\frac{\pi}{4} + \frac{(k-1)\pi}{6}\right) \sin\left(\frac{\pi}{4} + \frac{k\pi}{6}\right)}$ is equal to

[JEE (Advanced) 2016]

(A) $3 - \sqrt{3}$

(B) $2(3 - \sqrt{3})$

(C) $2(\sqrt{3} - 1)$

(D) $2(2 + \sqrt{3})$

MTR071

7. Let α and β be nonzero real numbers such that $2(\cos \beta - \cos \alpha) + \cos \alpha \cos \beta = 1$. Then which of the following is/are true? (where $\alpha, \beta \neq (2k+1)\pi$)

[JEE (Advanced) 2017]

(A) $\tan\left(\frac{\alpha}{2}\right) - \sqrt{3} \tan\left(\frac{\beta}{2}\right) = 0$

(B) $\sqrt{3} \tan\left(\frac{\alpha}{2}\right) - \tan\left(\frac{\beta}{2}\right) = 0$

(C) $\tan\left(\frac{\alpha}{2}\right) + \sqrt{3} \tan\left(\frac{\beta}{2}\right) = 0$

(D) $\sqrt{3} \tan\left(\frac{\alpha}{2}\right) + \tan\left(\frac{\beta}{2}\right) = 0$

MTR072

8. Let α and β be real numbers such that $-\frac{\pi}{4} < \beta < 0 < \alpha < \frac{\pi}{4}$. If $\sin(\alpha + \beta) = \frac{1}{3}$ and $\cos(\alpha - \beta) = \frac{2}{3}$, then the greatest integer less than or equal to $\left(\frac{\sin \alpha}{\cos \beta} + \frac{\cos \beta}{\sin \alpha} + \frac{\cos \alpha}{\sin \beta} + \frac{\sin \beta}{\cos \alpha} \right)^2$ is _____.

[JEE (Advanced) 2022]

MTR073

9. Let $\frac{\pi}{2} < x < \pi$ be such that $\cot x = \frac{-5}{\sqrt{11}}$. Then

$$\left(\sin \frac{11x}{2} \right) (\sin 6x - \cos 6x) + \left(\cos \frac{11x}{2} \right) (\sin 6x + \cos 6x)$$

is equal to :

[JEE (Advanced) 2024]

(A) $\frac{\sqrt{11}-1}{2\sqrt{3}}$

(B) $\frac{\sqrt{11}+1}{2\sqrt{3}}$

(C) $\frac{\sqrt{11}+1}{3\sqrt{2}}$

(D) $\frac{\sqrt{11}-1}{3\sqrt{2}}$

MTR130

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	A	C	B	B	B	A	D	C	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	D	D	B	D	C	D	D	A	B

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B,C	B,C	A,B	C,D	A,B,C,D	A,B,C,D	A,D	B,D	C,D	A
Que.	11	12	13	14						
Ans.	C	B	A	A→S; B→R; C→Q; D→P						

EXERCISE (S)

1.	0	2.	0	3.	1	4.	1.73	5.	5
6.	4	7.	5.75	8.	1.33	9.	150	10.	3
11.	9	12.	24	13.	11	14.	2	15.	2

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	3	4	1	2	3	4	2	1	2
Que.	11	12	13	14	15	16	17	18	19	
Ans.	80	4	1	1	1	48	2	1	1	

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	
Ans.	B	A,B	C,D	2	D	C	A,C	1	B	

Important Notes